



Degrowth in global spaces: Extractivism and practices of socio-ecological transformation in the geography of Chile's global fruit trade

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ABSTRACT

The current global socio-ecological crisis, caused by a growth-centered capitalist economy that disregards ecological limits and social injustices, underscores the necessity of rethinking human spatial relations beyond growth across multiple scales. Through a case study of Chilean fruit production for export to global markets, this research illuminates how spatialities of degrowth – a project of socio-ecological transformation beyond economic growth – can be conceived by examining the interplay between global, local, and intermediate scales. During extended field research, I identified different forms of production within the Chilean fruit industry: a wholly export-oriented agro-industrial and extractivist model coexists with a set of alternative practices regarding the modes of production (agroecology), trade (fair trade), and relations with the territories (supporting diverse local economies). While these different forms of production are intricately interconnected, and the alternative practices remain quantitatively marginal and incomplete, the latter points toward a socio-ecological transformation that may be defined as a project of solidary degrowth spaces. A transformation that acknowledges limits to growth, consequently, limits to international trade, yet operates without the illusion of eliminating global connections altogether, focusing instead on ways to organize them in fairer and more sustainable forms.

1. Global geographies of trade and Degrowth

Global trade relations, particularly concerning primary resources and agricultural products, during the history of colonialization and globalization have linked colonized and colonizing places where wealth was accumulated. These connections have often consisted in socially unjust and ecologically destructive processes such as extractivism, unequal (ecological) exchange, externalization of costs and damages, depeasantization, the exploitation of workers and small farmers, landscape operationalization, and ecological destruction, as well as the ecological impacts of logistics, while unequal power relations between involved actors, grant disproportionate power to multinational companies (see, for instance: [Arboleda, 2020](#); [Brand et al., 2016](#); [Hornborg, 2006](#); [Moore, 2000](#); [Topik et al., 2006](#)). These processes have been both fundamental drivers and consequences of economic growth. In this paper I specifically employ the concept of extractivism, originating in Latin American debates. I adopt a broad understanding of extractivism as the extraction of natural resources or primary agrarian products from dependent territories at high social and ecological costs, with (often transnational) corporations controlling critical points of trade, and typically characterized by unequal exchange ([Félic, 2024](#)). In other

words, “extractivism is based on socio-ecologically destructive processes of subjugation, depletion, and non-reciprocal relations” ([Chagnon et al., 2022](#): 762).

Meanwhile, research within the framework of de- and post-growth demonstrates that infinite economic growth is structurally incompatible with ecological sustainability as a main cause of the global socio-ecological crisis ([Hickel and Kallis, 2020](#); [Parrique et al., 2019](#)). Degrowth, as a consequence, posits that only a selective, democratically-led, and equitable reduction of production and consumption can serve as the foundational condition for a post-capitalist socio-ecological transformation that enhances social justice and well-being at the global scale, accompanied by redistributive measures. This requires a cultural change, especially in the Global North, to conceive new ways of well-being, that rely more on social relations, arts, culture etc., and less on economic success and wealth ([D'Alisa et al., 2015](#); [Demaria et al., 2013](#); [Kallis et al., 2020](#)).

Based on these two arguments – that (1) economic growth at all costs drives the socio-ecological crisis and that (2) a selective reduction of production and consumption is needed – many scholars examining pathways to spatialize degrowth, have proposed relocalizing economies and politics ([Krähmer, 2022](#); [Mocca, 2020](#)). However, is a binary

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approach of ‘local good’ versus ‘global bad’ adequate for addressing the multidimensional ecological crisis and social injustices? I propose a differentiated approach that, while taking seriously the critiques of globalization and global trade, takes into equally serious consideration the work of those who have investigated the complex relationality of space and place, to avoid the “local trap” (Purcell, 2006) of attributing certain positive or negative characteristics *a priori* to a certain scale. I refer, for instance, to Doreen Massey’s work (Massey, 2005), her recognition of the local making of the global, as well as the global making of the local, of the complex entanglements of space and place which cannot simply be disentangled. Instead, she argues, we should consider the “local responsibility for the global”, and we need context-dependent negotiations between degrees of openness and closure within these scales (ibid.). Similarly, the debate on planetary urbanization and relative city-hinterland relations (Brenner and Schmid, 2015; Brenner and Katsikis, 2020) illustrates the highly problematic subjugation of hinterlands as operational landscapes of capitalism but at the same time shows the degree to which places, as a matter of fact, are constructed through global spatial relations.

These complex realities of the global interconnectedness of place and space render proposals of a simple negation and abolition of such global relations implausible. Moreover, some degree of trade can contribute to resilience in the face of the uncertainties of climate and ecological crisis (if this does not result in excessive dependencies for food security) and can also serve as an instrument for promoting diversified, less meat-based nutrition in regions with climatic conditions that impede sufficient agricultural production, such as Patagonia, Northern Europe or the Sahel Zone (Fader et al., 2013). Furthermore, Porkka et al. (2013) demonstrate that food trade has historically improved food availability in many countries but has done so in a highly unequal way, benefiting wealthier countries far more than poorer ones. Thus, even in a degrowth perspective, which argues for a just and selective reduction of production and consumption (Chertkovskaya et al., 2019; D’Alisa et al., 2015; Demaria et al., 2013), global trade should be considered through a differentiated lens. In relation to Latin American extractivism, Féliz (2024), argues that “degrowth could only be achieved by ‘delinking’ the economy from the international process of value-formation”, also dependent on acts of international solidarity from the consumption side. A differentiated degrowth perspective on global trade geographies should advocate for a *quantitative* limitation and/or reduction of traded volumes (a *tendency* of relocation), while at the same time developing strategies to *qualitatively* transform such trade relations from extractivist to solidary, i.e. relations of exchange built for mutual benefit and not based on profit maximization (Brand, 2020; Eckardt, 2020; Krähmer, 2022; Krähmer and Brokow-Loga, 2024; Krähmer and Cristiano, 2022).

In this article, I explore these hypotheses through the case of Chilean production of fruit for export, chosen as an example of a global trade relation. The in-depth analysis of this case (in the tradition of studies of commodity geographies, chains, circuits, networks: Bair and Werner, 2011; Bakker and Bridge, 2006; Coe et al., 2008; Hughes and Reimer, 2004; Knowles, 2014; Leslie and Reimer, 1999; Topik et al., 2006) has discerned the mechanisms, actors, power relations that shape such a geography and its social and ecological impacts. The case study has contributed to understanding how and why a global trade relation becomes unsustainable and unjust in connection to the dynamics of economic growth. Simultaneously, it has allowed to differentiate between the specific causes of injustice and unsustainability and the global scale of the relation: the global scale is related to but is not identical to the causes of injustice and unsustainability. These, rather, are rooted in imbalanced power relations between actors and the extractivist character of a model of production and trade that aspires to limitless growth, viewing export as a panacea for growth. The presence of alternative practices in the context of this global trade relation has enabled a concrete reflection on the possibilities of solidary degrowth spaces across multiple scales, without rejecting the global scale as such.

This article proceeds as follows: after discussing materials, methods and positionality (Section 2), I focus on the destructive realities of the dominant extractivist model of fruit production in Chile (Section 3). In Section 4 I discuss alternative practices. The concluding section reflects on how these results can contribute to a conceptualization of solidary degrowth spaces at the global scale.

2. Materials, Methods and Positionality

The geography of Chilean fruit production and export encompasses a complex web of relations between a diverse array of places and actors. Chile today is the main fruit exporter of the southern hemisphere. The country has exported fruit for centuries, but only with the process of agrarian reform and counter-reform in the 1970 s, with strong state support, has growth accelerated significantly. Today, this industry supports thousands of farmers and workers, spanning large stretches of Chilean territory from the center-north to the center-south of the country. Consequently, a key methodological challenge was studying such a complex geography in-depth (see also Valz Gris, 2021). While the research addresses a global geography, the focus was on the region of production, particularly the impacts of global trade on producing territories. Most empirical research was conducted in Chile, spanning December 2021 to March 2022, as part of my PhD dissertation (Krähmer, 2023), on which this article builds. Interviews (n = 87) were the primary research method, combined with fieldwork conducted within Chile. Work in Chile was complemented by a visit to the Macfrut Trade Fair in Rimini (September 2021, 19 interviews) and five interviews in Europe, mainly Italy, to cover the buyer side. In Chile, I focused on several case studies of specific sites of production with associated trade connections, combining interviews with direct observation, and juxtaposing these with insights derived from the literature, statistical data, and grey literature, such as reports from state agencies, industry lobby organizations, and NGOs.

As shown in Fig. 1, fieldwork concentrated on central Chile, where the majority of fruit production occurs, focusing on four main case studies, indicated by yellow circles in the map, as well as several other sites, denoted by smaller yellow dots. Detailed descriptive accounts of the case studies can be found in Krähmer (2023). Here I provide a short summary of the cases; in the following two sections I discuss the implications of their analysis. Of the main case studies, two represent the mainstream extractivist model (informing Section 3): the Leyda case regards a large-scale avocado plantation (three interviews: plantation manager, owner, export company, plus informal conversations with workers). In the Nancagua case, I studied a medium-sized plantation in a regional context heavily shaped by fruit production (four interviews: plantation manager, owner, export company, and informal conversations with workers). These sites were complemented by 26 ‘general’ interviews with individuals involved in the industry, including producers, traders, experts, public servants, or external observers, such as activists and academics. Many of these interviews occurred in Santiago de Chile, a sort of ‘operational center’ of this extractivist model (Ye et al., 2019, cited in Chagnon et al., 2022). Alternative practices (informing Section 4) were investigated in the case of the cooperative company Mifruta in the Aconcagua Valley, which produces grapes and raisins and cooperates with fair trade buyers (eleven interviews: small producers, technician, export company). Another case is the Mapuche¹ community of Tralcao which produces cherries for export while simultaneously building a strong local economy (thirteen interviews: small producers, export company, researcher, managers of protected nature site). Additionally, across three different sites – Leyda, Yumbel, and Pinto – I investigated agroecological practices (six interviews: producers, adviser, researcher). Interviews regarding alternatives were more numerous due to their collective nature, in contrast to larger companies controlled by

¹ The largest indigenous group in Chile.

Figure 1: Municipal Distribution of fruit production in Chile and locations of fieldwork

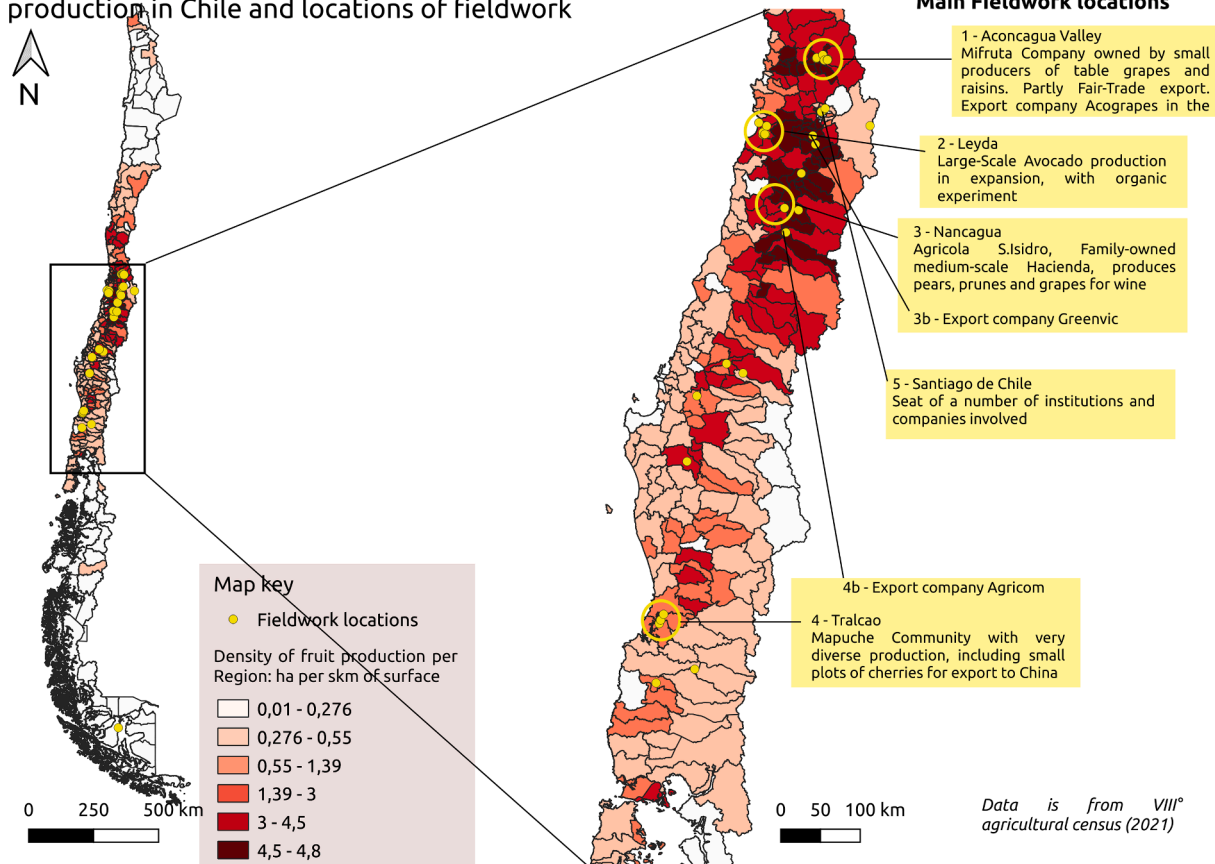


Fig. 1. Municipal distribution of fruit production in Chile and Fieldwork locations (by author).

owners. Conversely, the characterization of the mainstream model relied heavily on the ‘general’ interviews, while Chile lacks a comparable, comprehensive debate on alternatives. Indeed, what I propose here is an asymmetrical comparison: a model on one side, a collection of practices on the other. This asymmetry explains why sections 3 and 4 are not perfectly parallel and why, in Section 4, I place greater emphasis on the specific local experiences.

Throughout the research, interviewees were selected to encompass various perspectives, including those of producers, export companies, consultants, activists, academics, and state institutions. Frequently, one contact would lead to another. Interviews lasted between twenty minutes and an hour and a half and were semi-structured, combining common general questions with questions tailored to the interviewee’s specific role. There was also space for exploring themes emerging during conversations. Most interviews were conducted in Spanish; others were in English, Italian and German. While most interviews were conducted in-person, a minority was conducted online. According to the permissions granted via informed consent, the majority of interviews were recorded and transcribed; 49 interview transcriptions are accessible on Zenodo.² Only the excerpts used in the article were translated by me.

All collected materials were qualitatively analyzed as part of writing my PhD dissertation (Krähmer, 2023) over a period of about two years, progressively building the arguments I discuss here. Following the approach of case studies and the scope of social science as articulated by Flyvbjerg (2006) and Lancione (2013), I privileged logical and thematic connections between sources rather than systematic interview analysis. Issues raised in one interview informed questions in another or

led to the inclusion of previously overlooked dimensions or the pursuit of specific experts, statistics, documents, or academic literature.

My positionality in this research is characterized by my longstanding engagement with the degrowth framework, both as a researcher and activist. As Schoenberger (2007: 31) puts it, “[t]he politics help you identify the important questions, but they can’t tell you what the answers are”. I have tried to conduct research that contributes to this framework but in which I allowed the case study to evolve, remaining open to elements beyond or contrary to my initial assumptions. Another issue of positionality relates to me being a white male European middle-class researcher in Latin America. This research would have been impossible without the enormous support from those I worked with in Chile – including people working within the extractivist model that I critique. Chile is a relatively wealthy country and there is a risk of artificially distancing it by using the categories of Global North and Global South. Nonetheless, I emphasize that the conclusions I propose here about Chile benefit from an outsider’s perspective as much as they are constrained by it. These considerations are tied to the broader debate about degrowth not being an universalist answer to a universalist capitalist modernity, but instead as part of a pluriverse of alternatives beyond development and growth (Demaria and Kothari, 2017; Kothari et al., 2014; Nirmal and Rocheleau, 2019).

3. Territories as Operational Landscapes for Economic Growth: Extractivist Realities of Fruit Production in Central Chile

In this section, I first provide contextual background on the geographical distribution and the historical development of Chile’s fruit export industry. Subsequently, drawing on fieldwork and interviews, I analyze three main dimensions of ecological unsustainability and social

² <https://zenodo.org/record/7602356>

injustice that characterize this system as extractivist. Finally, I argue that these characteristics are linked to the global scale of Chile's fruit trade through six interdependent mechanisms.

The history of fruit production for export in Chile begins with small volumes and dried fruit as early as the 16th century (cfr. Fig. 2), integrated into the Spanish colonial system with its large-scale landownership and feudal structures. Through a process of 'ecological imperialism', European agricultural techniques were introduced in Chile (Lacoste et al., 2011; Silva and Marin, 2016). Chilean history, in general, has been shaped by the export of numerous commodities, and this economic model focused on commodity exports, persisted after independence in 1818 (Sagredo Baeza, 2014; Bustos-Gallardo and Prieto, 2019). The modern version of this model, on which I focus here, involving larger export volumes and a focus on fresh fruit, emerged at the beginning of the 20th century (Murray, 2002; Robles, 2010; Robles and Kay, 2018). Throughout this development, the Chilean state played a fundamental coordinating role. It attracted knowledge – particularly from California, aiming to replicate the fruit export model developed there – and organized subsidies and supporting state organizations. Crucially, the process of agrarian reform and counter-reform between the 1950s and the 1970s transformed a feudal, low-productivity agrarian system into a highly productive entrepreneurial system of farms (ibid.). The actual boom of Chilean fruit production for export began in the 1980s, marking a phase of exponential growth that continued at least until 2019. This expansion occurred within a global context of increasing international trade in fresh fruit and vegetables (Friedland, 2004; Huang, 2004).³ The Chilean export boom is thus not a specifically Chilean phenomenon but a component of a global trend towards increased market exchange of fruit and vegetables, territorialized in Chile. However, Chile's participation in this process has not been incidental: it has been guided by decisive policy interventions, enacted collaboratively between the state and the fruit industry. These interventions aligned with Chile's broader economic growth strategy under the neo-liberal regime of Pinochet's dictatorship. As a result, Chile has been characterized as a country of "commodity-regions" (Bustos-Gallardo and Prieto, 2019; Daher, 2003) and as a "planetary mine" (Arboleda, 2020). Today, fruit is the second-most exported group of commodities within the Chilean economy. Chile has become one of the world's leading fruit-exporting countries and is the top exporter in the Southern Hemisphere. These countries export fruit based on the 'counter-seasonality' of production: fruit ripens in the opposite part of the year in the Southern Hemisphere compared to the Northern Hemisphere, allowing to ensure year-round supplies of fresh fruit of the same species.

Today, large parts of the Chilean Central Valley, both north and south of Santiago, along with some surrounding hilly areas, are dominated by fruit plantations. Between them are industrial processing plants operated by export companies, cities and villages and the country's central highway. Substantial tracts of land have been transformed – decades or centuries ago in some cases, more recently in others – into fruit plantations of varying sizes and dedicated to different fruit species. Together, these plantations form a monotonous landscape of monoculture.

The rest of this section is based on findings from fieldwork and interviews conducted in Chile (cfr. Section 2). The organization of the production and export of fruit represents a complex reality involving numerous actors (cfr. Fig. 3). While the production process contains elements of a linear commodity chain (cfr. Fig. 3a), this chain exists within a far more intricate network of relationships (cfr. Fig. 3b), specifically with the state and the broader environmental and social context.

Also in the 'linear' chain, there are a number of different routes fruit can take, because a large number of companies with very distinct

characteristics are involved. For instance, farms are of different sizes and types: from medium-sized farms with the owners living there (e.g., a 40 ha pear, prune, and grape farm in the Nancagua case study) to large farms owned by absentee landowners (e.g., a c. 1000 ha of avocado plantation in the Leyda case study, cfr. Fig. 4). Furthermore, there are 'alternative' practices, discussed in the following section.

Focusing on the dominant model of fruit production and export, notwithstanding their diverse characteristics in terms of size and ownership, these farms have many things in common. Based on the results of my empirical research, I describe this model as extractivist due to three characteristics evident in the territories of production that parallel Chagnon et al.'s (2022) definition of extractivism (cfr. Section 1): (1) an instrumental use of the territory subjugated to an operational landscape; (2) a treatment of natural resources (e.g., water) as if they were unlimited, resulting in environmental damage and distributional injustices and conflicts; (3) unequal power relations at the production end of the commodity chain that, among other effects, create precarious labor conditions. These characteristics of extractivism are related to the system's export orientation, mediated through six mechanisms that I will discuss below.

First, there is the instrumental use of the territory, treated as a 'passive surface' on which to ground a global economic system, territorialized in Chile by an economic-political strategy, i.e., a place lacking its own agency, subjugated to the goal of producing economic growth (cfr. Spanier and Feola, 2022). In the terminology of planetary urbanization (Brenner and Katsikis, 2020), these territories are reduced to "operational landscapes". In the empirical reality of Chile this means large areas dedicated to agricultural and economic monocultures for export production, with very low degrees of agrarian and biodiversity. Agrarian production for local and national markets is marginalized, and many staple foods, once produced locally, are imported. Nature and its inevitable variations are controlled as much as possible: trees are planted in orderly rows; technicians experiment with the right distances in order to maximize production, the flow of water is organized and regulated, pesticides are applied to reduce the risk of pests; and the moment of harvest is painstakingly programmed to the last detail in order to guarantee fruit ripe enough to attract the far-away consumer while unripe enough to endure several weeks-long trips on ships. A whole economic-political system of different actors organizes this system together as a sort of growth coalition (cfr. Fig. 3).

Second, natural resources are treated as if they were unlimited. In the case of water (fully privatized in Chile) this treatment, in combination with the decade-long climate-crisis-driven drought in Chile (Garreaud et al., 2020), has led on some occasions to extreme water shortages for local communities, e.g., in Petorca.⁴ While Petorca is an extreme case, it illustrates the character of the economic model in the face of crisis: the production of avocados is saved even through the illegal extraction of water, irrespective of the impact that this has on the local communities' water access. Furthermore, many interviewees in the industry emphasized technical solutions to increase the availability of water (desalination, water highways) or use it more efficiently (drip irrigation), arguing that these could easily solve the issue if there were sufficient state intervention. Production limits, however, were not deemed necessary, according to these interviewees. This hints at the risk of cases like Petorca repeating themselves when the combination of a growing production and persistent drought exhausts increases in availability and efficiency. An activist of MODATIMA,⁵ an organization that fights for a fairer distribution of and access to water, argued:

⁴ See, for instance: <https://cl.boell.org/es/2018/07/26/el-impacto-socio-ambiental-de-la-industria-de-paltas-en-la-provincia-de-petorca> (Last access: July 2022).

⁵ <https://modatima.cl/> (Last access: August 2022).

³ See also data available in FAOSTAT and relative graphs in Krähmer (2022).

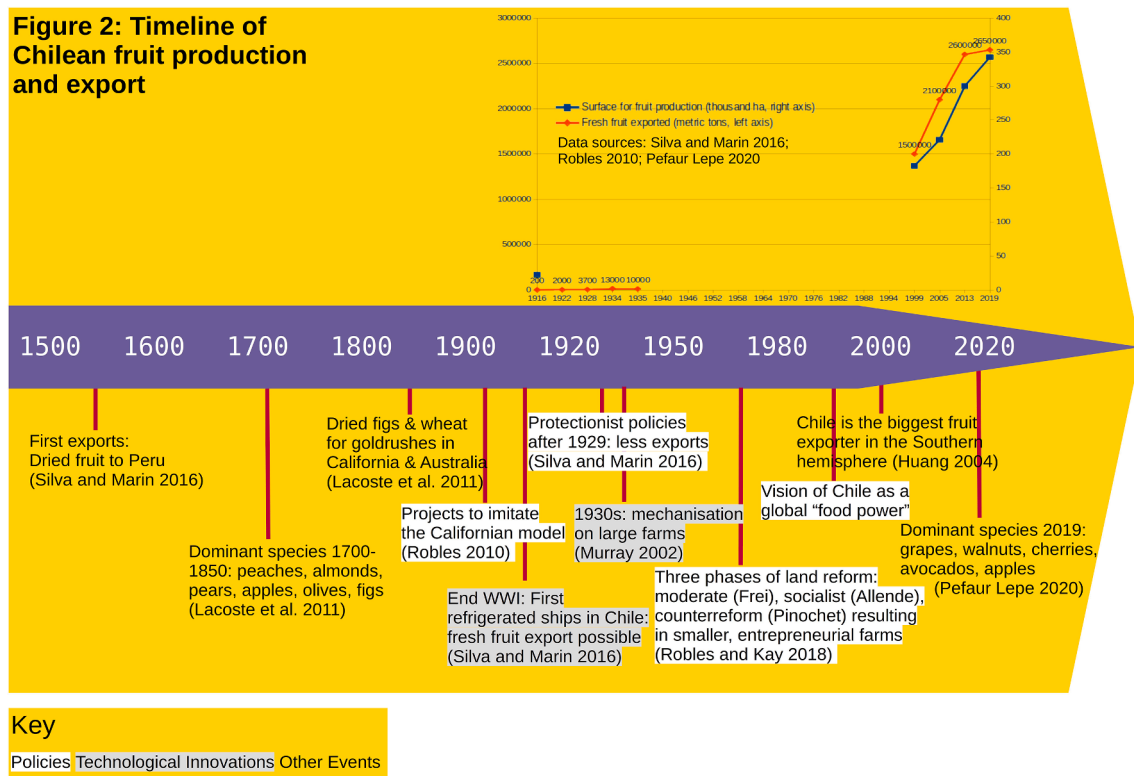


Fig. 2. Timeline of Chilean fruit production and export (by author).

When you make an irrigation system more efficient, what you do is that the person using this irrigation system will end up expanding more, because you can expand your cultivated areas, being more efficient. So finally, expanding the cultivated surfaces, they will end up consuming more water (Interview to Carlos Gutierrez).

This closely parallels the Jevon's paradox or rebound effect (Parrique et al., 2019), often cited in degrowth literature, that efficiency gains are offset by further growth of production. Of course, water resources are not unlimited, especially in the context of an accelerating climate crisis, and actors in the industry presumably are aware of this. But acting as if these limits could be indefinitely pushed further, without limiting production at a certain point, they act as if there were no limits and put at risk the long-term sustainability of fruit production itself. In a context of social inequalities, furthermore, the water availability for local communities is undermined in the process. While this is most evident in the case of water, fertile land and biodiversity are similarly exploited, oblivious of limits to growth.

Third, unequal power relations at the production side of the chain contribute to precarious labor conditions for both employed workers and smallholders farming their land directly. Most workers are temporarily employed (Baquero and Klein, 2012) but in some cases, one harvest after another, they work up to eleven months per year, traveling from region to region, from farm to farm. Workers often suffer from heat, sun and exposure to pesticides and are badly paid.⁶ The workforce in harvest and fruit processing is predominantly migrant and female (Baquero and Klein, 2012). "It's to survive, nothing more. (...) It's the only thing we are able to do. (...) There is no opportunity to do anything else", two plantation workers commented in an interview. Work is often organized through intermediaries who help secure jobs but deduct fees from

salaries; social security is rarely paid for – sometimes because the workers themselves, due to a lack of trust in the state and the pension system, refuse contracts, as Alicia Muñoz, activist of rural and indigenous women's organization ANAMURI⁷ explained, and interviewed workers confirmed. Workers find themselves at the lower end of a chain of unequal power relations. Farmers themselves are often unable to improve conditions, at least smaller and medium-sized farmers, as this quote by the manager of export company Acograpes illustrates (related to the Mifruta case study):

The big producers, due to economies of scale, can survive and arrange with other scales of sales or trade, but for a medium-sized or small producer, it is harder to be competitive, due to the costs they have (Interview to Cristian Ferriere).

Small farmers in particular often depend on a single export company, while the opposite is rarely the case. Only large producers generally have more bargaining power to negotiate the economic conditions of their sale.

These characteristics – or negative impacts – of the extractivist Chilean fruit industry in the territories of production are not caused by fruit export and its global geographies of trade directly but are instead mediated through six interrelated mechanisms: (1) Long-distance transport; (2) Standardization and quality requirements; (3) Technology and knowledge; (4) Specialization; (5) Unequal power relations in the global commodity chain; (6) Export as a mechanism of growth.

(1) Global export relies on global transport. It is worth noting, that, in food, transport itself (as long as it occurs by ship, as is the case for the majority of Chilean fruit) is only a minor source of GHG emissions⁸ (Crippa et al., 2021). However, global transport has relevant implications for the process of production. It requires careful selection of the

⁶ Around 20–30 thousand Chilean pesos for a full day of work were reported, that was at the time of the study less than 20–30 euros, or around 20–30 dollars in a country with costs of living not dissimilar from Europe.

⁷ <https://www.anamuri.cl/> (Last access: July 2022).

⁸ Much more relevant is the type of foodstuff consumed: 'local' meat usually has a much greater climate impact than 'global' fruit or vegetables.

3a) Simplified, linear chain

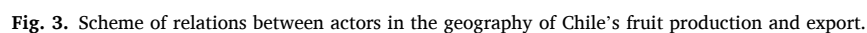
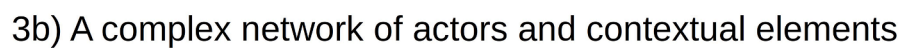




Fig. 4. New avocado plantation in Leyda 'under construction' (Photography by author).

harvest time to collect fruit capable of surviving long trips, as well as careful selection of the fruit. Packaging must be done with care, transport must be refrigerated and constantly monitored, and transport is a significant economic cost, requiring organization and planning along the entire chain to ensure fruit has the characteristics necessary to achieve sufficiently high prices in the destination market.

(2) Fruit must meet a number of quality requirements and certifications, generally imposed by importing countries and companies and controlled by certifications rules defined far away (GlobalGAP is the standard but often other types of certifications and quality requirements are requested), and verified by certification agencies. While these requirements and certifications are intended to have positive social and ecological implications, many interviewees were skeptical of their actual positive impact in Chile, and controls often focus on paperwork, rather than controlling on-the-ground conditions. Quality requirements are inextricably linked to a standardization process: at least five steps of selection and control regarding size, color, and sugar content occur between production on the field and the treatment of the fruit in the packaging plants of the export companies in order to conform fruit to the varying preferences and the fashions of the respective export market. The goal is to produce perfect and, most importantly, homogeneous, fruit, as required for a product destined to appear on a supermarket shelf, as this quote by a manager of export company Agricom (related to the Tralcao case study) illustrates:

every tree is different but in the packing process you try to homogenize these parameters (...). You try to leave a product that is so consistent in the box that it nearly seems artificial. (...) every person opening the product should meet a characteristic quality, perhaps not the best nor the worst but consistent, always the same apple in the same package, so that it is easier to repeat the buy. (Interview to Pablo Badilla).

(3) Both transport and quality requirements necessitate the use of a large amount of technology and knowledge. For this reason, many experts and consultants are active in Chile's fruit industry, directly employed by large firms, or by private consultancies and public agencies. Their knowledge has a cost and accessing this knowledge (whether about specific methods of fruit cultivation or market price

trends) gives larger actors (larger producers, export companies, etc.) an advantage over smaller ones, further contributing to power imbalances. Furthermore, certain types of knowledge are more valued than others in this context: it matters more to know about market trends in Europe than to have a detailed knowledge of the local ecosystem.

(4) The complexity of participation in global export markets in various ways favors specialization. High investments – not only in a plantation itself but also in knowledge – make specialization attractive: firms must be reasonably sure that investments will produce returns. This supports plantations organized as monocultures, an important feature of operationalized landscapes, as discussed above. Monocultures here refer not only to larger plantations but also to regional monocultures, where smaller plantations form patchworks growing the same or similar products. Many interviewees acknowledged the risks of specialization and often agreed on the need for diversification. However, most interpreted this as a need to diversify markets, showing much less interest in agrarian or biodiversity.

(5) Unequal power relations between actors along the chain – already discussed above as a characteristic of the extractivist model in Chile – extend beyond the Chilean border and persist throughout the global commodity chain, mediated by the laws of value and competition, knowledge disparities and personal relations. While profits in some parts remain with local landowners, a large share flows to Santiago de Chile (where many absentee landowners and the headquarters of many larger export companies are located) or further away to multinational export companies. Export companies compete with others in Chile and countries like South Africa and New Zealand. Among the most powerful actors are supermarket chains in the USA, Europe, and China. These final buyers in many ways define the conditions under which production and trade operate. They determine many of the rules, quality requirements, and prices.

(6) Export itself can be interpreted as a model of growth: producers' or state strategies to export, often influenced by buyers' demand, are fundamentally motivated by the desire to grow, to find new markets and to be able to sell more and/or at higher prices. Without export there would be no market for all the fruit produced in Chile – and there would not be so much fruit production. In this sense, in a degrowth perspective, production for export can be seen as intrinsically problematic.

Nevertheless, also among interviewees in the fruit industry the question of further future growth was debated controversially. An extreme pro-growth position was the following, certainly not coincidentally proposed by a manager of the fruit producer lobby organization Fedefruta:

I believe that Chile should hold a much more radical position in relation to the development of the agrarian production and the food processing sector. It should consider this a more powerful industry (...). Chile should concentrate on an industry with a lot of force, and the way to do this is to industrialize its agriculture (Interview to Mario Marin Valdebenito).

In contrast, Catalina Cuevas from the state pro-export agency Pro-Chile argued that having built a fruit export economy around a perspective of unlimited growth has been a fundamental problem, leading to shortsighted strategies ignorant of natural limits and cycles.

In response to some of the unsustainable impacts of Chile's fruit production (water, climate impacts), a number of initiatives are promoted by the industry and the state (a detailed discussion can be found in Krähler, 2024). Most of them though appear to be flawed in the typical manner of "sustainability fixes" or "green fixes" (Holgersen and Malm, 2015; While et al., 2004). These aim to enhance the fruit's appeal to 'conscious' consumers in the Global North, rather than bringing about real change in the modes of production. They operate primarily on a technological level, avoiding structural changes. These initiatives under the generic banner of 'sustainability' appear to be a far cry from the requests of activists who focus instead on specific issues, such as the decommodification of water⁹ or improving labor conditions.¹⁰

In sum, Chile's export-oriented fruit industry can be described as extractivist, in that it maintains an instrumental relation with the places where it operates, utilizing them as operational landscapes or passive surfaces to produce economic growth. This approach leads to multiple social and ecological impacts, closely linked to the global export economy through a number of mechanisms. Sustainability initiatives that avoid structural change are bound to fail. Nevertheless, the fruit export industry remains a significant economic factor for Chile, particularly in rural areas, providing (albeit precarious) work to many, especially those facing difficulties in finding alternative employment such as migrants, rural women, or students. Indeed, conversations with workers revealed not only complaints about working conditions but also expressions of general satisfaction with their work. Many workers feared of a crisis of the export model. Recognizing this tension underscores the importance of a nuanced perspective on Chilean fruit production for global trade, not limiting the analysis to its negative sides but also considering potential trajectories of change, anticipated in existing alternative practices.

4. Practices for a Socio-Ecological Transformation of Global Trade?

Not all fruit production for export in Chile follows an extractivist logic. In this section I discuss learnings from case studies of alternative approaches to organizing production and trade in Chile (cfr. Section 2), to conceptualize a different idea of production and trade beyond a dichotomous distinction between 'bad' global trade and 'good' local production. In this conceptualization, global trade is quantitatively limited but exists, organized differently. These practices can be defined as alternatives as they attempt to resolve at least one of the three problematic characteristics of the mainstream extractivist model: (1) they do not use the territory in a purely instrumental manner, avoiding operationalization; (2) they attempt to use natural resources aware of their limits; (3) they create more equal power relations and better labor conditions. However, there are two limits to the analysis presented here:

first, these cases are marginal in quantitative terms¹¹; second, these alternatives do not form one cohesive alternative model but rather each of them is an alternative practice in relation to parts of the problematic characteristics of the mainstream system discussed in Section 3.

I have studied three cases of alternative practices (see also Fig. 1): First, the Mapuche community of Tralcao (near Valdivia) has built strong local economies instead of viewing export as a panacea. In this case the territory is far from an operational landscape; rather, it is the habitat of a lively community that cares for it holistically. Furthermore, they are extremely aware of limits to growth. Second, agroecological practices, studied in three sites, take particular care of natural resources, for example, favoring biodiversity in their plantations. Third, the cooperative company Mifruta (in the Aconcagua Valley) focuses on the cooperation of small producers and is a good example of how power relations between producers, the export company and buyers can be shaped more equally thanks to cooperation and fair trade. Instead of discussing these case studies one after another (detailed information on each case can be found in Krähler, 2023), I focus here on their interconnections and on how, in combination, these small practices might contribute to the development of an alternative model.

The socio-ecological impacts characteristic of the extractivist model are linked to global trade mediated through a set of mechanisms (see Section 3). In analogy, the alternative practices discussed here are connected with global trade through different mechanisms. While they share with the mainstream model the implications of long-distance transport (1), each of these practices includes at least some of the alternative mechanisms of (2) building collaborative quality requirements; (3) using technology differently and valuing other forms of knowledge; (4) favoring biodiversity and specializing to a lesser degree; (5) creating solidary relations; (6) accepting limits to growth and redefining export as a contribution to resilience and diversification.

(1) Long-distance transport remains a challenge for these practices. It requires levels of organization that are difficult to achieve at small scales. Mifruta has partly eased this challenge by specializing in dried fruit, raisins, which are easier to store and transport. In agroecological production, a paradox arises: organic blueberries, among the few fruits that achieve high enough prices, are sometimes flown by plane from Chile to the USA at the beginning and the end of the season, resulting in a far worse carbon footprint than fruit transported by ship (Crippa et al., 2021; FIA and INIA, 2010). Producers interviewed in Pinto are aware of this contradiction and they plan to invest more into products for the Chilean market. This reminds of the tension of organic fruit being more sustainable in production but related to elite consumption (Guthman, 2004).

(2) Quality requirements take on a different dynamic in the case of Mifruta: in the context of fair trade they are developed in dialogue with producers – defining for instance together what the fair trade bonus will be used for – and they are more oriented at guaranteeing social (e.g., labor) and ecological standards, instead of focusing on standardization for consumers. Similar arguments can be made in relation to agroecological certifications. This different mechanism has its limitations as fair trade and organic products remain niche markets and Mifruta can only sell a part of their production through fair trade channels. Due to insufficient demand for fair trade products, the farmers of Tralcao have not managed to access this channel and continue to sell on the mainstream market. Meeting export standards and quality requirements poses significant challenges for them, in terms of technical knowledge and investment capacity, due to their small scale (each producer in the community grows between 0.5 and 2 ha of cherries).

(3) The role of technology and knowledge shifts in complex ways between the mainstream model and the alternatives. In agroecological

⁹ See MODATIMA, <https://modatima.cl/>. (Last access: July 2022).

¹⁰ See ANAMURI, <https://www.anamuri.cl/>. (Last access: July 2022).

¹¹ No precise number on their relevance is available, as they do not fall under any specific category identifiable in official statistics but it was not easy to find them and interviewees agreed that they were exceptions to the norm.

practice, technology is employed less for the scope of standardization – for example, no pesticides are used. Agroecology employs different agricultural techniques, promoting biodiversity. In the case of Mifruta, improved power relations with exporters are also achieved through better access to knowledge: as a cooperative business, Mifruta's small producers can pool resources to employ a professional technician. They have also expanded their geographies, with some traveling to England to meet their clients – this may sound banal but in relation to the extended travel geographies of someone like the much wealthier owner of the plantation in Nancagua, this is an important step towards the reduction of structural inequalities between producers of different sizes. Furthermore, alternative practices value different kinds of knowledge: instead of a knowledge instrumental to the industrial logic of the export industry, they more often emphasize contextual knowledge about biodiversity and local ecosystems for agroecology, about the relation to their land in a holistic, respectful and also spiritual way in Tralcao.

(4) The prioritization of agrarian and biodiversity replaces monocultures and specialization. In agroecology, this is an essential element: biodiversity helps to reduce agrochemicals. But different positions have emerged among interviewees if 'real' agroecology is related to the scale of production and if there are differences between a 'simple' organic farming and a form of production that incorporates biodiversity at its core. For instance, a large, organically certified avocado plantation in Leyda – owned by the same family as the one mentioned in Section 3 – does not differ much in the structure of the plantation and its relation to the territory from its 'conventional' counterpart. In contrast, a small blueberry plantation in Pinto is the habitat of a large diversity of flowers and other spontaneous plants, and dedicates space to small biotopes (Fig. 5). In Tralcao, rather than transforming their territory into a passive surface for economic growth, the community of small Mapuche farmers has actively cultivated a diversified local economy. This approach is deeply tied to their history and geography, as Tralcao is located in a river ecosystem.

The resistance against the contamination of this environment by a large Arauco paper mill has been a uniting force in and beyond the Mapuche community of Tralcao. This mill is part of the extractivist forest industry, which dominates the surrounding region with its monoculture of forest plantations that contrasts sharply with the territory of Tralcao, which is extremely diversified and colorful. Farmers in Tralcao consciously chose to cultivate only small portions of their land for export, retaining most of their space for local production and self-consumption. Besides cherries for export, they produce other fruit, breed animals, tend vegetable orchards, and invest in small-scale eco-tourism projects on the river. They produce mainly for the local market which they organize through direct contacts to clients and an annual village fair (Fig. 6), where they sell cherries and derivatives, alongside cakes and juice, meat and other food, accompanied by handwork, music, and dance, in close contact with the nearby city, Valdivia. Mifruta, by contrast, operates in a largely monocultural context – a valley long dedicated to the production of grapes – but they promote some efforts of diversification, such as producing goat cheese.

(5) Solidary relations are established and unequal power relations are challenged by the cooperative and fair trade practices in the case of the small grape producers of Mifruta. Through cooperation, they have achieved better conditions within the fruit export economy. Although Mifruta is formally a company, not a cooperative,¹² it operates with democratic and solidary structures. It has effectively improved its position within the power dynamics of the export economy, as their president argues (and many members confirmed) when comparing their situation before and after founding the company:

We didn't have power in buying and selling. (...) Now we can analyze if a client really is convenient for us or not. We can obtain higher prices and we have obtained them. If we had acted individually, like in the beginning, many producers would have disappeared because it is a lot of effort to fight alone (Interview to Pedro Lucero).

For Mifruta, alongside the personal dedication of members, fair trade has played an important role. Even though only a small portion of their overall production is sold under fair trade, the stable prices and fair trade bonuses they receive enable planning, support for local social projects, and assistance for producers in crisis. Nevertheless, many face threats from the reduced and increasingly scarce water availability in their valley, exacerbated by growing avocado plantations on the surrounding hills. By contrast, power relations remain problematic in Tralcao, where relations with export companies are managed by small single farmers. The community's initial cooperative structure collapsed, and relationships with export companies are perceived as unjust, often resulting in insufficient payments or even no payment at all when costs along the chain exceed revenues. Unequal power relations remain an issue, as one community member explains:

We have exported with five different companies. With dramatically bad results. (...) There is always a quite good year and the following year things go worse (Interview to Eduardo Hernández).

This situation, aggravated by the lack of a collective organization to secure better conditions, has led several of producers to abandon export production altogether.

(6) Finally, it is more nuanced to assess if export remains interpreted as a mechanism of growth in these cases. To some degree, it is, but without ambitions for *limitless* growth. There is a clear awareness of limits: the agroecological blueberry producers in Pinto planned to reduce exports by plane, even though profitable, due to their high ecological impact. Farmers in Tralcao wanted to improve their economic condition thanks to fruit export but they had a clear growth limit in mind in the respect of their common territory. They never conceived planting cherries for export mainly as a strategy for growth but rather as a means of economic diversification, as community leader Pedro Guerra says:

All that is export or fruit, should be a part for export, another part that should continue with wheat, or in the north livestock breeding (...). It is an alternative but it's not the idea of a fruit production to export only. Also production for the national market and also production of other things (Interview to Pedro Guerra Huechante).

Export production is integrated into a rich and diversified local landscape and economy, actively built and protected by the community and their networks. Community members are deeply aware of this connection, rooted in their attachment to their land, grounded in the Mapuche cosmovision.

This analysis has shown how the mechanisms linking fruit production in Chile to export differ in these alternative practices compared to the mainstream extractivist industry, as well as the limitations of these alternatives. Only if all these mechanisms of relating territories of production to global trade were a fully implemented and cohesive reality, one could speak of a true alternative model. However, these alternative practices do not exist in isolation from the extractivist model described in Section 3. Both are deeply intertwined through the laws of value (the flow of money), quality requirements and changing tastes (influencing demand and prices from the consumer side), and unequal power relations often tied to unequal access to knowledge, in which small producers are in a structurally weaker position than large export companies or supermarket chains, considering the type of knowledge valued in global markets. Furthermore, from the moment of sale to the export companies – in relation to which small producers are in a weak position – the fruit enters the same dynamics of the wider industry, if not sold fair trade or organically certified, with the large power and the strict quality requirements of supermarket chains. However, critically, these

¹² Cooperatives in Chile were dissolved under the Pinochet dictatorship and until today they often suffer from a bad reputation and are now rare exceptions in Chilean agriculture.



Fig. 5. On the agroecological blueberry farm in Pinto close to Chillán (Photography by author).



Fig. 6. Tralcao's summer village fair (Photography by author).

alternatives do not transform territories into operational landscapes for economic growth, as they are enabled by strong agencies of the actors and communities who promote them. Without their efforts, the different mechanisms described here would not exist. This agency is certainly much more locally situated than that of the state and the big and multinational companies of the extractivist fruit industry.

5. Conclusions: towards solidary degrowth spaces

The study of the Chilean fruit export industry in its dominant extractivist form supports critiques of global commodity geographies as

unsustainable and unjust (Brand et al., 2016; Cook et al. (2004); Hornborg, 2006; Moore, 2000; Topik et al., 2006). This extractivist model uses territories instrumentally, operationalizing them, treats natural resources as if they were unlimited, leading to distributional conflicts along lines of social injustices and creates imbalanced power relations with precarious labor conditions. At the same time, the study of alternative practices shows the potential for organizing production for global trade in more sustainable and solidary forms.

This comparative analysis avoids the 'local trap' (Purcell, 2006) of attributing a specific quality to a particular scale *a priori*. At the same time, it does not support indifference to scale. Instead, it adopts a

differentiated perspective on the complex entanglements between local, global, and intermediate scales, considering the scale of economic activities and actors alongside geographical scales. Indeed, while the negative characteristics of the extractivist model of fruit production for export is certainly related to global trade, it is so mediated through six mechanisms (cfr. Section 3 and Fig. 7) in which diverse actors in Chile (producers, exporters, diverse articulations of the state) play central roles. These mechanisms constitute the causes for the unjust and unsustainable reality of this geography of global trade. Changing these mechanisms (cfr. Section 4 and Fig. 7), alternative practices – enabled by localized agencies – can redefine the relation between local production and global export markets. But for this transformation, the scale of economic activity is critical, particularly in terms of power relations and of economic growth. Regarding power relations, interactions between large multinational companies and small producers create imbalances that cooperation among producers can partially alleviate but which also raises the problem of economic concentration and capital accumulation. Regarding economic growth, the proposed *qualitative transformation* of global trade relations can only be conceived in the context of a *quantitative limitation* and/or reduction of total trade volumes. The ecological impacts of transport, infrastructures, and packaging demand such a limitation. More importantly, the alternative practices discussed embed production for export within healthy local economies. This implies the production of lower volumes for export, and more for local markets, than in the extractivist model, representing a sort of 'territorial limit of growth'. Only by respecting this limit can exporting serve as a means of diversification contributing to economic resilience, rather than operationalizing landscapes. For instance, the high diversity of productions in Tralcao, is central for its non-instrumental use of the territory. Unlimited growth of fruit production for export – even if done by the same community – would render this diversity impossible. This is consistent with the theoretical argument from the introduction, which understands relocalization in a degrowth perspective as a tendency toward quantitative limitation, combined with the need to build different, solidary relations, also at the global scale. These alternative practices cannot be easily replicated in other contexts. A strong community like in Tralcao or in the case of Mifruta, or the knowledge about agroecology are enabling conditions which are not universally present, especially in territories long operationalized for monocultural export-oriented production, like many areas in Chile. Moreover, the 'technological lock-in effects' of

extractivism (Chagnon et al., 2022) – the physical operationalization of a territory for an extractivist economy – hinders transformations towards alternative models. Thus, processes of disentangling places from an extractivist system of production and trade are complex and must be differentiated across diverse geographies. For these practices to coalesce into a cohesive alternative model, their diverse mechanisms must be integrated. While no single mechanism is entirely transformative, their combination, this is the hypothesis, can effect meaningful change. Moreover, such an alternative model cannot rely solely on isolated, strongly motivated actors. As with the extractivist model, strong transversal coalitions (albeit with different characteristics) are needed. The state could play a central role in such coalition, when aware of its responsibility of governing territories for ecological sustainability and social justice.

The conclusions presented here are based on research conducted in Chile from an external viewpoint, employing a degrowth framework that does not claim to provide universal solutions (cfr. Sections 1 and 2). They are not intended as *prescriptions* for Chile. I would be glad, of course, if they somehow contributed to a Chilean debate, whether within the degrowth framework or others, such as post-extractivism or *buen vivir* (Kothari et al., 2019). At the same time, these are *lessons from* Chile, made possible thanks to the support I received there, for an idea of degrowth in global spaces. Within this attention to context, epistemologies, and positionality, three key dimensions of change emerge as central to a socio-ecological transformation of global trade that contributes to solidary spaces of degrowth: (i) the transformation of production (favoring biodiversity over specialization and monoculture, valuing not only instrumental forms of knowledge); (ii) the fairness of trade, beyond existing fair trade niches (creating collaborative quality requirements and building solidary relations); (iii) the integration of production for export within diversified local economies (recognizing limits to growth and viewing export not as a panacea but as a contribution to economic diversification and resilience of local economies and of food systems globally). In this research, I have focused on production territories and how they link to global trade. Further studies could examine the consumption side and the dynamics of trade itself, exploring further pathways of transformation.

This research illustrates the complex relationality of global space, characterized by complex webs of interdependencies rather than (only) linear dynamics of exploitation. The places studied in Chile are

Comparative table of the extractivist model and alternative practices		
	Extractivist Model	Alternative Practices
Characteristics in the territory of production	Landscape Operationalization	Non-instrumental use of the territory
	Natural resources treated as unlimited	Awareness of limits
	Unequal power relations (in the territory of production)	More equal power relations (in the territory of production)
Mechanisms that mediate the connection with global trade	Global Transport	
	Standardization & quality requirements	Collaborative quality requirements
	Technology & knowledge	Different use of technology, valuing other forms of knowledge
	Specialization	Agrarian and biodiversity, less specialization
	Unequal power relations (in global commodity chains)	Solidary relations (in global commodity chains)
	Export for growth	Export for resilience and diversification

Fig. 7. Comparative table of the extractivist model and alternative practices.

collections of stories, in which power is unequally distributed, they are “articulations within the wider power-geometries of space” (Massey, 2005). However, agency is distributed, and the case cannot be reduced to a simple dualism. Neither is this a case of global trade as a “tide [that] lifts all boats” nor an inevitable “race to the bottom” (Werner, 2018). While alternative practices are in problematic ways entangled and interconnected with an extractivist industry, they offer hope for the project of solidary degrowth spaces, in which relations of exchange are not organized for the one-sided capitalist and growth-oriented exploitation of nature and human communities but for mutual benefit (Brand, 2020; Krähmer and Brokow-Loga, 2024).

CRedit authorship contribution statement

Karl Krähmer: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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